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Level overlap and level color coding revisited: Improved attribute attendance and higher choice consistency in discrete choice experiments

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ABSTRACT

Previous work has identified attribute level overlap and level color coding as effective and attractive strategies to reduce task complexity and improve behavioral efficiency in discrete choice experiments (DCEs). However, the simultaneous and combined impact of level overlap and level color coding on attribute non-attendance and choice consistency has not yet been investigated. To address this limitation and to strengthen the available evidence base, this paper re-analyzed an existing randomized controlled DCE from the Netherlands (N = 2,731) and analyzed a new randomized controlled DCE conducted in the United Kingdom (N = 3,084) using heteroskedastic attribute non-attendance mixed logit models. Both randomized controlled experiments were based on a relatively complex instrument with 5 attributes with 5 levels each and the results from both experiments were remarkably similar. In the base-case study arms without level overlap and color coding, only about half of the attributes are attended to. Level color coding as a stand-alone strategy improves attribute attendance but reduces respondents' choice consistency. In contrast, level overlap as a stand-alone strategy improves attribute attendance while simultaneously increasing respondents' choice consistency. The combination of level overlap and color coding is even more effective: it results in approximately full attribute attendance and a 30% increase in respondents' choice consistency. Experimental designs with level overlap are therefore recommended as a default design strategy and level color coding recommended to further increase respondents' behavioral efficiency in complex DCEs.

1. Introduction

In discrete choice experiments (DCEs), researchers face an inherent trade-off between statistical and behavioral efficiency (Jonker et al., 2019). On the one hand, higher statistical efficiency yields more informative choice tasks that result, assuming that the respondents' preference structure remains constant, in smaller sample size requirements (Rose and Bliemer, 2013; De Bekker-Grob et al., 2015). On the other hand, more statistically efficient choice tasks tend to increase the level of task complexity and reduce respondents' choice consistency (e.g., DeShazo and Fermo, 2002; Caussade et al., 2005; Louviere et al., 2008). The latter corresponds to smaller absolute preference coefficients cq. a smaller effect size, which creates an opposite effect that increases sample size requirements.

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Higher task complexity also tends to induce simplifying choice behavior such as attribute non-attendance (e.g., Hensher 2006; Grebitus et al., 2018; Veldwijk et al., 2023). This results in the seemingly paradoxical observation that imposing design constraints to reduce the level of task complexity may not only improve behavioral efficiency but can potentially also reduce sample size requirements in DCEs, irrespective of the specific efficiency criterion that is used; see e.g. Kessels et al. (2006) for a comparison of different efficiency criteria.

Previous studies have introduced attribute level overlap as an appealing and straight-forward approach to constrain design efficiency and enhance behavioral efficiency in DCEs (e.g., Maddala et al., 2003; Johnson et al., 2010; Jonker et al., 2018, 2019; Meyerhoff and Oehlmann, 2023). Attribute level overlap involves fixing the levels of one or more attributes across different choice tasks at the same value, making the tasks more manageable while encouraging respondents to consider all attributes during the decision-making process (Johnson et al., 2010; Kessels et al., 2015). This not only has a positive impact on respondents' behavioral efficiency, but also allows practitioners to obtain preference information for the less or even least important attributes in the choice process (Jonker et al., 2019).

The visual layout of choice tasks is another aspect of DCEs that can be used to decrease the level of task complexity for respondents. For example, DCE designs that include level overlap can be presented in a partial profile layout, in which the overlapped levels are hidden and only the non-overlapped levels are shown to respondents. [1] This reduces the visual complexity of the choice tasks and has been used to accommodate a larger number of attributes in DCEs (e.g., Chrzan, 2010). However, partial profile layouts are not as commonly used anymore because they can result in omitted attribute bias and learning effects resulting from respondents trying to impute the missing levels in the partial profiles (Bradlow et al., 2004). Moreover, partial profile layouts are incorrect in the presence of interaction effects between the overlapped (hidden) and non-overlapped (visible) levels. Consequently, most published DCEs are nowadays based on full profile presentations of the choice tasks, in which both the overlapped and non-overlapped levels are visible.

With full profile choice tasks, a range of visual strategies can be used to reduce the complexity of the choice tasks. For example, the visual presentation of the choice tasks can be used to help respondents to more easily identify the levels that are overlapped, for instance by highlighting the levels that are not overlapped. This can be done via different background colors (e.g., Norman et al., 2016, 2019; Mulhern et al., 2018; Rowen et al., 2021; Nicolet et al., 2022), differential font colors (e.g., Luyten et al., 2022; Assele et al., 2023), or, when the attribute levels are strictly ordinal, with intensity color coding schemes such as traffic light colors or color shades (e.g., Jonker et al., 2017, 2018; Goossens et al., 2019; Janssen et al., 2020; Himmler et al., 2021, 2022; Dieteren et al., 2023). Thus far, several studies have demonstrated that color coding, such as traffic-light colors, shades of purple, or highlighting of differences, are effective visual aids that can improve clarity, reduce task complexity and improve choice consistency in DCEs (e.g. Jonker et al., 2018, 2019; Assele et al., 2023).

Despite the existing evidence that both attribute level overlap and level color coding are able to improve various aspects of respondents' behavioral efficiency, the simultaneous and combined impact of level overlap and level color coding on attribute non-attendance and choice consistency has thus far not been investigated. Only separate analyses have been published that either take attribute non-attendance or choice consistency into account. To address this limitation and to strengthen the available evidence, this paper aims to re-analyze an existing DCE and analyze a new DCE using heteroskedastic mixed logit models that simultaneously incorporates attribute non-attendance and choice consistency effects. Based on the presented results, the paper aims to encourage practitioners to use level overlap, possibly combined with color coding, to improve behavioral efficiency of respondents in their DCEs.

2. Methods

2.1. Two randomized discrete choice experiments

Two randomized controlled discrete choice experiments with five and six experimental study arms, respectively, were used to investigate the separate and combined impact of level overlap and color coding on respondents' choice consistency and attribute non-attendance. The different experimental study arms were created by systematically combining experimental designs with and without level overlap with different types of color coding (see Tables 2a and 2b).

The first randomized discrete choice experiment was conducted in the Netherlands and the choice data were previously described and analyzed by Jonker et al. (2018, 2019). In short, a DCE with 21 choice tasks per respondent was created based on the EQ-5D-5L instrument (Herdman et al., 2011). The EQ-5D-5L is a generic health state instrument that comprises five dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension has 5 levels: no problems, slight problems, moderate problems, severe problems and extreme problems. EQ-5D-5L health states are defined by selecting one of the five levels from each dimension and respondents were asked to repeatedly choose their preferred state out of two EQ-5D-5L health states with fixed 10-year life spans (see Fig. 1).

The second discrete choice experiment was a separate study conducted in the United Kingdon. The setup and design of the UK study mimicked that of the Dutch experiment, although with the highlighting of differences color coding scheme replaced by a traffic light color coding scheme and with six instead of five study arms (see Fig. 2). Results from this study have not previously been published.

[1] DCE designs with level overlap constraints are historically referred to as 'partial profile designs' because they were specifically optimized for partial profile layouts (Kessels et al., 2011). However, partial profile layouts are not commonly used anymore, which implies that 'partial profile designs' are usually presented in a full profile layout in which the overlapped levels cannot be removed from the design. Hence there is typically 'partial' about partial-profile DCE designs, and therefore 'level overlap designs' provides a more informative description of DCE designs with level overlap constraints.

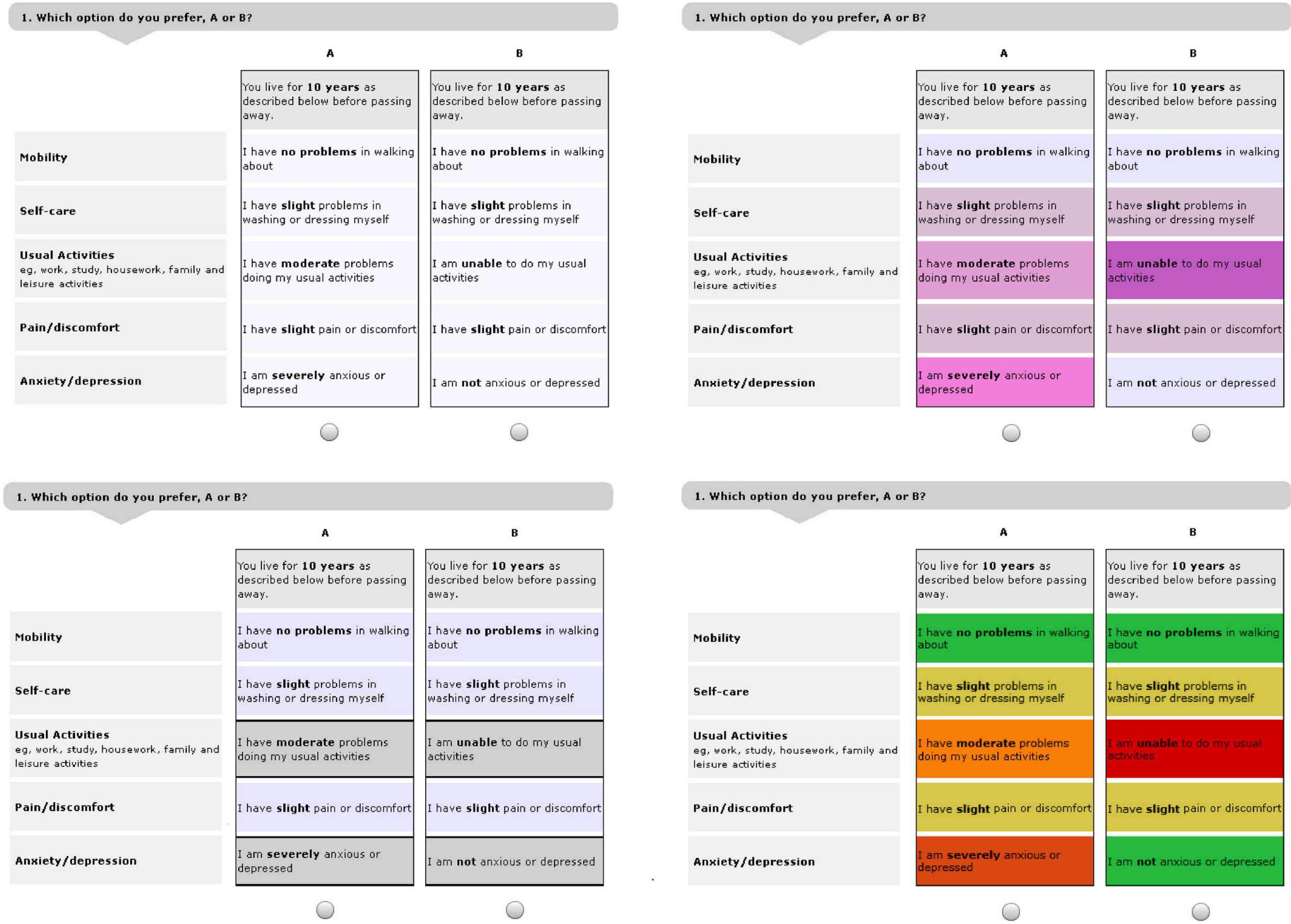

Fig. 1. Visual presentation of the choice tasks*
*Note: larger screenshots are included in the online supplemental.

2.2. DCE designs

For both randomized controlled experiments, Bayesian D-efficient design algorithms were used to optimize two DCE designs: one without and one with level overlap. The designs without level overlap presented all attributes at different levels in each choice task. Conversely, the designs with level overlap were constrained to always present three out of the five included attributes at the same level. This created maximum statistical power to detect statistically significant differences in attribute non-attendance and choice consistency due to differences in attribute level overlap.

So-called heterogeneous DCE designs were used to improve the D-efficiency of the designs while allowing them to be more robust to respondent heterogeneity (Sándor and Wedel, 2005). Heterogeneous DCE designs consist of multiple subdesigns that are optimized concurrently. They are computationally more demanding to create, yet once created, each respondent is only asked to complete a single subdesign. Consequently, heterogeneous DCE designs do not require any additional effort from participating respondents.

The DCE design optimizations started from an initial design with 8 subdesigns and 21 random choice tasks per subdesign. Subsequently, a row-based Modified Fedorov algorithm (Fedorov, 1972; Cook and Nachtsheim, 1980) was used to search for more efficient designs. The search continued for 24 h to ensure near-optimal DCE designs given the specified level overlap constraints. The level overlap constraints were implemented by restricting the candidate set to the subset of permissible choice tasks. Hence for the designs without level overlap, a candidate set of all non-dominant choice tasks without any level overlap was specified. And for the designs with level overlap, a candidate set of all non-dominant choice tasks with 3 levels overlap was used.

As previously described in Jonker et al. (2018, 2019), the design optimization algorithms were implemented in Julia (Bezanson et al., 2012). The design optimization criteria were calculated as the weighted average Bayesian D-error with $^1/_3$ of the weight allocated to the Bayesian D-error of the overall DCE design and with $^2/_3$ of the weight assigned to the average of the Bayesian D-errors of the stand-alone subdesigns. The Bayesian D-errors were computed from the asymptotic variance-covariance matrix of a conditional logit model with linear-additive main-effects utility function, making use of a Latin hypercube sample (LHS) of 100 draws and informative priors. The latter were obtained from previous EQ-5D-5L research and updated after a pilot run of 200 respondents, and for the UK study again after 400 respondents had completed the survey, to maximize statistical efficiency.

2.3. Color coding

Both DCE designs (i.e. the designs with and without level overlap) were presented to respondents with and without color coding. Fig. 1 contains examples of the different layouts that were used. As shown, the base-case layout without color coding relied solely on textual emphasis through partial bolding to highlight the different levels. In addition, both the Dutch and UK experiments included a shades of purple intensity color coding scheme. This color coding scheme was originally created by researchers from the LISS panel (Scherpenzeel, 2011) for a health-state valuation study (Jonker et al., 2017). The shades of purple are optimized to indicate differences in attribute levels for respondents with red-green color vision deficiency, which is the most prevalent form of color blindness, while ensuring that the text remains legible for participants affected by other types of color vision impairments. In addition to shades of purple color coding, the Dutch experiment included a highlighting of differences color coding approach, which requires a DCE design with level overlap, whereas the UK experiment included a traffic-light intensity color coding approach in which the color red was used to denote the worst and green the best possible EQ-5D-5L levels.

2.4. Respondent recruitment and survey structure

Online, nationally representative samples from the Dutch and UK general populations in terms of age, sex, education, and for the Dutch sample, also geographic region were recruited via the same commercial survey sample provider, ie. Survey Sampling International. The target sample size exceeded 500 respondents per study arm to ensure ample statistical power for detecting differences between the study arms. In each study, participating survey respondents were randomly assigned to 1 of the study arms and they received a small financial compensation (i.e. approximately 1 euro in the Netherlands and 1 pound sterling in the UK) upon completion of the survey.

The Dutch and UK surveys were similarly structured. First, the surveys were briefly introduced. Next, respondents had to complete

A. Study arms in the Dutch discrete choice experiment

	Experimental design	
Color coding	without level overlap	with level overlap
no color coding	**1**	**3**
shades of purple	**2**	**4**
highlighting of differences	**n/a ***	**5**

B. Study arms in the UK discrete choice experiment

	Experimental design	
Color coding	without level overlap	with level overlap
no color coding	**1**	**4**
shades of purple	**2**	**5**
traffic light colors	**3**	**6**

** n/a = not applicable (i.e., highlighting of differences color coding requires a DCE design with level overlap)*

Fig. 2. Overview of the Dutch and UK randomized controlled experiments

a self-evaluation task in which they had to rate their current health in terms of the EQ-5D-5L instrument. This step was designed to acquaint respondents with the format and visual presentation of the health states. After this, two warm-up choice tasks were included to introduce respondents to the layout and type of trade-offs involved in the DCE. After the warm-up tasks, the primary set of 21 choice tasks were presented to respondents in a random order. To prevent excessive repetition, a few background questions were interspersed after the 7th and 14th choice task. Finally, at the end of the survey, respondents were required to complete several debriefing questions.

2.5. Statistical analyses

Respondents' health state preferences were modeled using a heteroskedastic mixed logit model with stochastic attribute selection. The stochastic attribute selection (see e.g. Scarpa et al., 2009) was used to account for attribute non-attendance and to estimate the average number of attributes attended by respondents in each study arm. In addition, different scale parameters were included to estimate the relative choice consistency of the study arms and the impact of level overlap and color coding on attribute and level importances, respectively.

More formally, let $X_{i,t,j,d,k} \in \{0, 1\}$ denote whether level k of attribute d is present in choice option j in choice task t of respondent i and let L_d denote the number of levels that attribute d contains. The implemented utility function is then:

$$U_{i,t,j} = \sum_{d=1}^{D} \sum_{k=1}^{L_d-1} \beta_{i,d,k} * \tau_{i,d} * \lambda_a * \varphi_d * \psi_{c,k} * X_{i,t,j,d,k+1} + \varepsilon_{i,t,j} \tag{1}$$

in which the beta-parameters (β) capture the utility decrements of each non-base case level, the binary tau (τ) parameters reflect whether attribute d was attended to by respondent i, the lambda (λ) parameters capture scale differentials between the study arms (with a denoting the study arm in which respondent i participated), the phi (φ) parameters reflect the relative impact of level overlap on the importance of the included attributes, the psi (ψ) parameters capture the relative impact of the different types of color coding, indexed by c, on the importance of the attribute levels, and the error term (ϵ) is assumed to be independent and identically Gumbel distributed.

The estimation of eq. (1) crucially depends on the systematic variation of level overlap and color coding that was introduced via the different study arms of the randomized controlled experiments (see Fig. 2). In addition, to ensure statistical identification, the following constraints were imposed on the scale parameters.

- First, the lambda parameters were defined to measure the choice consistency relative to the base-case study arm without level overlap and color coding, which means that $\lambda_1 \equiv 1$.
- Second, the phi parameters measure attribute importance effects of level overlap relative to the study arms without level overlap, which means that $\varphi_{1:D} \equiv 1$ for all study arms without level overlap. In addition, to ensure that different attribute importances do not confound the other scale and beta parameters, a mean-of-one constraint was imposed on the phi parameters for the study arms that did include level overlap, ie. $\frac{\sum_{d=1}^{D} \varphi_d}{D} \equiv 1$. This means that the other parameters in the utility function are on average multiplied by 1.0 and thus remain unaffected by differential attribute importance effects.
- Third, the psi parameters measure level importance effects of the different types of color coding relative to the study arms without color coding, which means that $\psi_{1:L_d-1} \equiv 1$ was imposed for all study arms that did not include color coding. In addition, independent mean-of-one constraints were imposed on the psi parameters of each color coding type, ie. $\frac{\sum_{k=1}^{L_d-1} \psi_k}{L_d-1} \equiv 1$ for all c. As before, this ensures that the other parameters in the utility function are on average multiplied by 1.0 and consequently unaffected by eventual level importance effects.

Conform standard MIXL assumptions, the respondent-specific β-parameters were assumed multivariate normal distributed with mean μ and variance-covariance matrix Σ:

$$\beta_i \sim \text{Multivariate Normal}(\mu, \Sigma), \tag{2}$$

where β_i denotes a vector of all $\beta_{i,d,k}$ parameters for respondent i. During model estimation, the average attribute importance, which is the population mean preference estimate (μ) of the most important attribute level of each attribute expressed as a percentage of the sum of the population mean preference estimates of the most important attribute levels, was monitored and subsequently reported in the comparison between study arms. Finally, similar to Jonker et al. (2018), the binary attribute attendance (τ) parameters were assumed independently Bernoulli distributed

$$\tau_{i,d} \sim \mathit{Bernoulli}(p_{d,a}) \tag{3}$$

with p denoting the probability of success, indexed by attribute d and study arm a. During model estimation, the average number of attributes attended to by respondents in each study arm was monitored and reported in the comparison between study arms.

If an attribute is not attended to by a respondent, the preference and scale parameters associated with that attribute are multiplied by 0, which implies that the attribute no longer affects the utility of this respondent. Conversely, the inclusion of the attribute attendance parameters implies that the estimated preference and scale parameters are derived solely from the responses of respondents who attended to the respective attributes.

Bayesian methods were used to estimate the MIXL models, which were implemented in the BUGS language and fitted using OpenBUGS software, making use of user-written extensions to increase the efficiency of the Markov Chain Monte Carlo (MCMC) sampling and implement the mean-of-one constraints on the scale parameters. The online supplemental contains the model codes and specification of the priors, including additional technical details about the sampling strategy that was used to enforce the mean-of-one constraints.

All models were estimated using 25,000 burn-in draws to let 3 different MCMC chains converge, and an additional 75,000 draws without thinning to reliably approximate the posterior distributions of the parameters and monitored quantities (ie, the number of attributes attended and base-case attribute importances). Convergence of the chains was evaluated based on a visual inspection of the chains and the diagnostics as implemented in the OpenBUGS software.

3. Results

Table 1 provides the descriptive statistics of the Dutch and UK randomized discrete choice experiments. As shown, the datasets are based on 2,731 and 3,084 respondents, respectively, with similar age and educational attainment. The major difference between the datasets is the drop-out rate during the DCE, which is 11% in the Dutch and 3% in the UK dataset. Note that more detailed descriptive statistics are included in the online supplemental.

Table 2a and b presents the posterior summary estimates of the attribute attendance and choice consistency effects. Both datasets provide very similar and consistent results, although the absolute level of attribute attendance and choice consistency is better in the UK than in the Dutch sample. In both datasets, attribute attendance is lowest in the study arms without level overlap and color coding. When color coding is added (as a stand-alone strategy), it improves attribute attendance but deteriorates choice consistency. In contrast, when level overlap is used as a stand-alone strategy, it improves both attribute attendance and choice consistency. The best results are obtained in the study arms that combine level overlap and color coding; when level overlap is combined with color coding, attribute attendance is higher and without negative effect on the choice consistency (with highlighting of differences) or with substantially higher choice consistency (with intensity color coding). Interestingly, the difference between the two types of intensity color coding is small but traffic light color coding appears to result in slightly higher attribute attendance than shades of purple.

Table 3 contains the posterior summary estimates of the absolute and relative attribute importance estimates in the Dutch and UK experiments. Starting with the base-case attribute importance estimates, anxiety/depression (attribute 5) is the most important attribute and selfcare (attribute 2) and usual activities (attribute 3) are the least important in the study arms without level overlap in the Dutch and UK experiments, respectively. Turning to the relative scale parameters, which measure the extent to which attribute level overlap affects the base-case attribute importance estimates, in both experiments a similar effect can be observed: the most important attribute becomes less important and the least important attribute becomes more important when level overlap is included in the DCE design.

Table 4 presents the level importance scale estimates of the different types of color coding schemes that were included in the study arms. Similar to the relative attribute importance scale effects, these scale effects measure the extent to which the various color coding schemes affect the level importance structure in the DCE. As shown, in both the Dutch and UK experiments, the introduction of shades of purple results in an increased importance of the 3rd level ('moderate health problems') compared to the 2nd level ('slight health problems') but without affecting the relative importance of the least important (2nd) and most important (5th) level of each attribute. The highlighting of differences color coding scheme, which was only included in the Dutch experiment, results in a more significant shift in the level importances, with the worst levels receiving increasingly more attention than the less important levels. The traffic light color coding scheme, which was only included in the UK experiment, behaves similarly to the shades of purple color coding scheme but with the distinction that the most extreme levels, which are indicated by the dark red colors, become significantly more important.

Table 1
Summary descriptives, by dataset [a].

Dataset	Variable	Completes					Dropouts (during DCE)				
		N	mean	stdev	min	max	N	mean	stdev	min	max
The Netherlands	age	2,731	48	17	18	98	296	56	16	18	89
	male		0.48		0	1		0.70		0	1
	educ_low		0.32		0	1		0.54		0	1
	educ_med		0.42		0	1		0.23		0	1
	educ_high		0.26		0	1		0.23		0	1
United Kingdom	age	3,084	46	18	16	89	103	52	20	16	84
	male		0.49		0	1		0.55		0	1
	educ_low		0.34		0	1		0.48		0	1
	educ_med		0.38		0	1		0.31		0	1
	educ_high		0.28		0	1		0.21		0	1

[a] Note: more detailed summary descriptives broken down by study arm are included in the online supplemental I.

Table 2a
Attribute attendance and choice consistency estimates [*] (Dutch dataset, N=2,731 respondents).

Study arm	1	2	0	3	4	5
Sample size per arm	541	540	0	553	546	551
Level overlap	No	No	No	Yes	Yes	Yes
Color coding	No	Shades of purple	Highlighting [b]	No	Shades of purple	Highlighting
Attribute attendance [0–5]	2.20 (1.93–2.47)	2.97 (2.65–3.25)	n/a	3.85 (3.62–4.08)	4.33 (4.17–4.49)	4.09 (3.90–4.31)
Relative scale (λ)	1.00 (reference)	0.86 (0.75–0.97)	n/a	1.12 (0.97–1.27)	1.32 (1.14–1.50)	1.09 (0.94–1.26)

[*] Note: mean posterior estimates with 95% credible intervals in parenthesis.
[b] Highlighting of differences requires level overlap in the experimental design.

Table 2b
Attribute attendance and choice consistency estimates [*] (UK dataset, N=3,084 respondents).

Study arm	1	2	3	4	5	6
Sample size per arm	544	512	513	501	510	504
Level overlap	No	No	No	Yes	Yes	Yes
Color coding	No	Shades of purple	Traffic light	No	Shades of purple	Traffic light
Attribute attendance [0–5]	2.68 (2.35–3.00)	3.62 (3.34–3.88)	4.14 (3.89–4.36)	4.24 (4.03–4.45)	4.51 (4.37–4.64)	4.64 (4.50–4.77)
Relative scale (λ)	1.00 (reference)	0.90 (0.80–1.02)	0.86 (0.75–0.97)	1.13 (0.99–1.28)	1.37 (1.20–1.55)	1.34 (1.17–1.52)

[*] Note: mean posterior estimates with 95% credible intervals in parenthesis.

Table 3
The effect of level overlap on relative attribute importance [a].

		Dutch dataset (N=2,731 respondents)	UK dataset (N=3,084 respondents)
attribute importance (%) in study arms without level overlap	attribute 1. mobility	0.18 (0.17–0.20)	0.18 (0.17–0.20)
	attribute 2. self-care	0.12 (0.11–0.14)	0.17 (0.16–0.18)
	attribute 3. usual activities	0.15 (0.14–0.17)	0.13 (0.12–0.14)
	attribute 4. pain/discomfort	0.22 (0.21–0.24)	0.22 (0.20–0.23)
	attribute 5. anxiety/depression	0.32 (0.30–0.34)	0.30 (0.28–0.31)
relative attribute importance scale effects in study arms with level overlap	φ attribute 1. mobility	0.88 (0.81–0.95) [b]	1.03 (0.96–1.09)
	φ attribute 2. self-care	1.12 (1.01–1.24) [b]	1.05 (0.99–1.12)
	φ attribute 3. usual activities	1.07 (0.97–1.16)	1.20 (1.13–1.28) [b]
	φ attribute 4. pain/discomfort	1.07 (0.99–1.16)	0.92 (0.87–0.98) [b]
	φ attribute 5. anxiety/depression	0.87 (0.79–0.95) [b]	0.80 (0.73–0.87) [b]

[a] Note: mean posterior estimates with 95% credible intervals (CI) in parenthesis.
[b] 95% CI does not comprise 1.0.

Table 4
The effect of color coding on relative level importance [a].

		Dutch dataset (N=2,731 respondents)	UK dataset (N=3,084 respondents)
level importance effects (shades of purple)	ψ level 2. slight health problems	0.96 (0.89–1.04)	0.93 (0.86–1.01)
	ψ level 3. moderate health problems.	1.10 (1.05–1.15) [b]	1.06 (1.01–1.11) [b]
	ψ level 4. severe health problems	0.93 (0.90–0.97) [b]	0.97 (0.93–1.01)
	ψ level 5. extreme health problems	1.00 (0.96–1.05)	1.04 (0.99–1.09)
level importance effects (highlighting)	ψ level 2. slight health problems	0.83 (0.73–0.94) [b]	n/a
	ψ level 3. moderate health problems.	0.86 (0.79–0.93) [b]	n/a
	ψ level 4. severe health problems	1.14 (1.08–1.20) [b]	n/a
	ψ level 5. extreme health problems	1.17 (1.10–1.25) [b]	n/a
level importance effects (traffic light)	ψ level 2. slight health problems	n/a	0.92 (0.84–1.00)
	ψ level 3. moderate health problems.	n/a	1.02 (0.97–1.08)
	ψ level 4. severe health problems	n/a	0.97 (0.93–1.01)
	ψ level 5. extreme health problems	n/a	1.09 (1.04–1.14) [b]

[a] Note: mean posterior estimates with 95% credible intervals (CI) in parenthesis, relative to study arms without color coding.
[b] 95% CI does not comprise 1.0.

4. Discussion

4.1. Summary of results

The primary aim of this study was to investigate and evaluate the individual and combined impact of level overlap and color coding on attribute non-attendance and choice consistency in two randomized controlled discrete choice experiments. The re-analysis of a Dutch and new analysis of a UK dataset revealed remarkable consistencies despite the higher overall data quality that was observed in the UK sample. That is, the introduction of color coding as a stand-alone strategy effectively increased attribute attendance, albeit at the expense of somewhat lower choice consistency. The introduction of level overlap as a stand-alone strategy resulted in higher attribute attendance without the disadvantage of lower choice consistency. Yet the most effective approach was the combination of level overlap and color coding, which resulted in almost full attribute attendance and an approximately 30% increase in choice consistency in both datasets.

4.2. Comparison with existing research

The presented results are congruent with previously reported findings and show that there are clear advantages and few disadvantages to the use of level overlap in discrete choice experiments. Level overlap promotes attribute attendance and helps avoid dominant attribute strategies by ensuring that there are always choice tasks in the DCE design where the most important attribute is overlapped. This occasionally necessitates respondents to make choices based on other attributes, thereby decreasing the relative importance of the most important attribute and increasing the amount of information obtained on less important attributes (Johnson et al., 2010). Another important benefit of level overlap is that it can improve attribute attendance up to the point where relatively standard (as opposed to more complex attribute non-attendance models) are sufficient to analyze the data, which greatly simplifies the analysis and reporting of the DCE results. More importantly, shifts in attribute attendance directly affect attribute importances and can thus affect marginal rates of substitution between the included attributes (see e.g. Hensher and Rose, 2009; Meyerhoff and Oehlmann, 2023). Accordingly, level overlap can also be an effective design strategy to help avoid biased willingness to pay (WTP) and maximum acceptable risk (MAR) estimates from the outset, i.e. already during the DCE data collection as opposed to afterwards via more complex econometric analysis.

Color coding presents a more complex picture. On the one hand, color coding significantly improves attribute attendance, especially when combined with level overlap in the DCE design, which concurs with previous analyses (e.g., Jonker et al., 2018). However, when used as a stand-alone strategy, color coding improves attribute attendance at the cost of reduced choice consistency, which is a novel finding based on the introduced modelling approach. Only the combination of color coding and level overlap improves attribute attendance and choice consistency at the same time. In addition, different color coding schemes were found to have varying impact on relative level importances. For example, the use of traffic light color coding intensifies the attention paid to the worst levels that are displayed in red colors, relative to not using any color coding. Similarly, highlighting of differences results in increased attention to the most important levels. The extent to which this creates or solves a problem depends on the DCE, but in most applications practitioners should be cautious when including traffic light colors because the colors have independent meaning and can change the DCE results relative to not using color coding. Moreover, the more neutral shades of purple were confirmed to provide an attractive alternative to traffic light color coding. They are equally effective (in terms of increasing attribute attendance and improving respondents' choice consistency), do not distort the relative distance between the best and worst levels, yet provide extra clarity to respondents about the correct ordering of the levels. As shown in Table 4, the shades of purple help respondents to correctly differentiate between levels 2 and 3 (i.e. 'slight' versus 'moderate' health problems) and country-specific value sets often contain illogical preference reversals precisely for these levels when color coding is not used (see e.g., Pickard et al., 2019; Andrade et al., 2020; Welie et al., 2020; Finch et al., 2022).

4.3. Limitations and future research

While the investigation into the effects of level overlap and color coding on attribute non-attendance and choice consistency has confirmed the robustness of previously published results, it is important to acknowledge several limitations of the implemented modelling approach.

First, the implemented modelling approach comes with an important and inherent limitation, which is that preferences and scale are intrinsically confounded in discrete choice models (Ben-Akiva and Lerman, 1985). This implies that the choice consistency estimates in each study arm crucially relies on the distributional assumptions placed on respondents' preferences and cannot be guaranteed to uniquely reflect scale as opposed to preferences (Hess et al., 2013). Similarly, it cannot be guaranteed that the attribute non-attendance estimates reflect pure attribute non-attendance effects; even in a model with random preference heterogeneity it is likely that small (i.e., close to but not zero) preference parameters are mistaken for attribute non-attendance, and vise versa (Hess et al., 2013). However, as previously mentioned by Jonker et al. (2019), even though preferences, scale, and attribute non-attendance may indeed have been confounded in each study arm, as a result of the carefully constructed randomized controlled experiments, random assignment of respondents across study arms, and identical modelling approach across study arms, the confound should be approximately constant. Hence the substantial differences in choice consistency and attribute attendance are still guaranteed to originate to a large extent from the different experimental conditions of the study arms.

Second, the presented attribute attendance and relative scale estimates are derived from a base-case model that accommodates attribute importance effects and level importance effects. To verify the robustness of the presented results with respect to the included attribute and level importance scale parameters, the online supplemental provides additional attribute attendance and relative choice consistency estimates based on a) a simplified model in which attribute and level importance effects are entirely ignored, and b) a more complex model in which level overlap and color coding are allowed to both affect attribute and level importances. In both cases, the attribute attendance and relative scale effects are close to identical, both in the Dutch and in the UK data set.

Third, the presented attribute attendance and scale estimates were also not corrected for the potential impact of warm-up and fatigue effects. Based on previous results, there were warm-up effects in the Dutch dataset in the study arms that did not include level overlap, but no evidence of fatigue effects in any of the study arms (Jonker et al., 2019). To confirm the absence of fatigue effects in the UK sample, the online supplemental also presents model results that incorporate additional warm-up and fatigue effect parameters. Indeed, there were no fatigue effects in any of the study arms. Additionally, there were statistically significant warm-up effects in all study arms, both in the Netherlands and in the UK experiment, when DCE designs without level overlap were used. In contrast, no warm-up effects were found in any of the study arms that included level overlap. Accordingly, these analyses confirm that the higher task complexity of DCE designs without level overlap did not create fatigue effects (i.e. decreasing choice consistency during the course of the DCE), but rather resulted in a combination of attribute non-attendance heuristics, lower choice consistency, and significant learning effects at the beginning of the DCE.

Fourth, the transferability of results to other discrete choice experiments is certainly an issue. For example, intensity color coding schemes, such as the shades of purple and traffic light colors, are only useful for attributes that possess a logical ordering of their levels. Otherwise, the logical ordering as conveyed by the colors would result in a cognitive dissonance with the actual categorical level structure of the attribute. As a result, the benefit of traffic light and shades of purple color coding may be less pronounced in DCEs with fewer attributes with a logical level ordering. In contrast, the highlighting of differences color coding approach remains applicable irrespective of the attribute level structures in the DCE – although the highlighting format depends on the presence of level overlap in the experimental design whereas intensity color coding can also be used without level overlap.

Fifth, as mentioned by one of the reviewers, the highlighting color coding format can be used to emphasize the overlapped as well as non-overlapped levels. The reason for highlighting the non-overlapped levels in the Dutch randomized discrete choice experiment was that it intuitively made sense to draw respondents' attention to the differences that they need to evaluate in order to make a choice. The opposite highlighting format, highlighting non-overlapped levels, would make it less intuitive for respondents to focus on the differences between the choice options despite the fact that these differences and not the similarities between the choice options logically determine respondents' choices. Consequently, there is an argument to be made for the evaluation of the highlighting of differences color coding scheme, but further research is necessary to establish the relative performance and differential impact of a highlighting of similarities relative to a highlighting of differences color coding scheme.

Finally, with respect to level overlap, further research to establish the optimal amount of level overlap is still necessary. On the one hand, keeping everything else constant, level overlap reduces the statistical efficiency of a DCE design and increases sample size requirements. On the other hand, level overlap is an effective strategy to improve behavioral efficiency, which implies that the priors do not actually remain constant. For example, the increased choice consistency implies larger preference parameters and hence a positive effect on the required sample size. With less complex choice tasks it is often also possible to include a few additional choice tasks, especially when the choice tasks are presented in a small number of blocks with a few neutral questions in between to avoid fatigue effects due the repetitive nature of the choice tasks. This way minimum sample size requirements can actually decrease despite the initial negative effect of level overlap on statistical design efficiency and, depending on the complexity of the DCE, less than the maximum amount of level overlap may strike a better balance between statistical and behavioral efficiency.

4.4. Conclusion

The presented results confirm that there are substantial benefits of including level overlap in experimental DCE designs. Accordingly, level overlap can be recommended as a default design strategy aimed at reducing task complexity and improving behavioral efficiency in DCEs. Color coding is also beneficial, particularly when combined with level overlap. However, not all types of color coding are equally effective and some types introduce stronger distortionary effects than others. This does not imply that color coding should not be used, only that practitioners should take into account that color information can exert an independent and potentially undesirable effect on respondents' preferences. More neutral colors, like shades of purple that are also easier to interpret for respondents with the most prevalent form of color blindness, present a relatively safe choice.

Conflict of interest disclosures

None.

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Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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References

Andrade, L.F., Ludwig, K., Goni, J.M., Oppe, M., de Pouvourville, G., 2020. A French value set for the EQ-5D-5L. Pharmacoeconomics 38, 413–425. Apr.
Assele, S.Y., Meulders, M., Michiels, H., Flamant, N., Vandebroek, M., 2023. The effect of information provision and color coding in product labeling on the preference for meat substitutes. Sustainability 15 (22), 15688. Nov 7.
Ben-Akiva, M.E., Lerman, S.R., 1985. Discrete Choice Analysis: Theory and Application to Travel Demand. MIT press.
Bezanson, J., Karpinski, S., Shah, V.B., Edelman, A., 2012. Julia: a fast dynamic language for technical computing. arXiv preprint arXiv:1209.5145. Sep 24.
Bradlow, E.T., Hu, Y., Ho, T.H., 2004. A learning-based model for imputing missing levels in partial conjoint profiles. J. Market. Res. 41 (4), 369–381. Nov.
Caussade, S., de Dios Ortúzar, J., Rizzi, L.I., Hensher, D.A., 2005. Assessing the influence of design dimensions on stated choice experiment estimates. Transp. Res. Part B Methodol. 39 (7), 621–640. Aug 1.
Chrzan, K., 2010. Using partial profile choice experiments to handle large numbers of attributes. Int. J. Mark. Res. 52 (6), 827–840. Nov.
Cook, R.D., Nachtsheim, C.J., 1980. A comparison of algorithms for constructing exact D-optimal designs. Technometrics 22 (3), 315–324. http://www.jstor.org/stable/1268315.
de Bekker-Grob, E.W., Donkers, B., Jonker, M.F., Stolk, E.A., 2015. Sample size requirements for discrete-choice experiments in healthcare: a practical guide. Patient Patient Cent. Outcomes Res. 8, 373–384. Oct.
DeShazo, J.R., Fermo, G., 2002. Designing choice sets for stated preference methods: the effects of complexity on choice consistency. J. Environ. Econ. Manag. 44 (1), 123–143. Jul 1.
Dieteren, CM, Bonfrer, I, Brouwer, WB, Van Exel, J, 2023. Public preferences for policies promoting a healthy diet: a discrete choice experiment. Eur. J. Health Econ. 24 (9), 1429–1440.
Fedorov, V.V., 1972. Theory of optimal experiments. In: Probability and Mathematical Statistics, vol. 12. Academic press, New York (N.Y.).
Finch, A.P., Meregaglia, M., Ciani, O., Roudijk, B., Jommi, C., 2022. An EQ-5D-5L value set for Italy using videoconferencing interviews and feasibility of a new mode of administration. Soc. Sci. Med. 292, 114519. Jan 1.
Goossens, L.M., Jonker, M.F., Rutten-van Mölken, M.P., Boland, M.R., Slok, A.H., Salome, P.L., van Schayck, O.C., in t Veen, J.C., Stolk, E.A., Donkers, B., 2019. Research team that developed the ABC tool. The fold-in, fold-out design for DCE choice tasks: application to burden of disease. Med. Decis. Making 39 (4), 450–460. May.
Grebitus, C., Roosen, J., 2018. Influence of non-attendance on choices with varying complexity. Eur. J. Market. 52 (9/10), 2151–2172. Oct 12.
Hensher, D.A., 2006. How do respondents process stated choice experiments? Attribute consideration under varying information load. J. Appl. Econom. 21 (6), 861–878. Sep.
Hensher, D.A., Rose, J.M., 2009. Simplifying choice through attribute preservation or non-attendance: implications for willingness to pay. Transport. Res. E Logist. Transport. Rev. 45 (4), 583–590. Jul 1.
Herdman, M., Gudex, C., Lloyd, A., Janssen, M.F., Kind, P., Parkin, D., Bonsel, G., Badia, X., 2011. Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). Qual. Life Res. 20, 1727–1736. Dec.
Hess, S., Stathopoulos, A., Campbell, D., O'Neill, V., Caussade, S., 2013. It's not that I don't care, I just don't care very much: confounding between attribute non-attendance and taste heterogeneity. Transportation 40, 583–607. May.
Himmler, S., Soekhai, V., van Exel, J., Brouwer, W., 2021. What works better for preference elicitation among older people? Cognitive burden of discrete choice experiment and case 2 best-worst scaling in an online setting. J. Choice Model. 38, 100265. Mar 1.
Himmler, S., Jonker, M., van Krugten, F., Hackert, M., van Exel, J., Brouwer, W., 2022. Estimating an anchored utility tariff for the well-being of older people measure (WOOP) for The Netherlands. Soc. Sci. Med. 301, 114901. May 1.
Janssen, E.M., Dy, S.M., Meara, A.S., Kneuertz, P.J., Presley, C.J., Bridges, J.F., 2020. Analysis of patient preferences in lung cancer–estimating acceptable tradeoffs between treatment benefit and side effects. Patient Prefer. Adherence 3, 927–937. Jun.
Johnson, F.R., Ozdemir, S., Phillips, K.A., 2010. Effects of simplifying choice tasks on estimates of taste heterogeneity in stated-choice surveys. Soc. Sci. Med. 70 (2), 183–190. Jan 1.
Jonker, M.F., Attema, A.E., Donkers, B., Stolk, E.A., Versteegh, M.M., 2017. Are health state valuations from the general public biased? A test of health state reference dependency using self-assessed health and an efficient discrete choice experiment. Health Econ. 26 (12), 1534–1547. Dec.
Jonker, M.F., Donkers, B., de Bekker-Grob, E.W., Stolk, E.A., 2018. Effect of level overlap and color coding on attribute non-attendance in discrete choice experiments. Value Health 21 (7), 767–771. Jul 1.
Jonker, M.F., Donkers, B., de Bekker-Grob, E., Stolk, E.A., 2019. Attribute level overlap (and color coding) can reduce task complexity, improve choice consistency, and decrease the dropout rate in discrete choice experiments. Health Econ. 28 (3), 350–363. Mar.
Kessels, R., Goos, P., Vandebroek, M., 2006. A comparison of criteria to design efficient choice experiments. J. Market. Res. 43 (3), 409–419. Aug.
Kessels, R., Jones, B., Goos, P., 2011. Bayesian optimal designs for discrete choice experiments with partial profiles. J. Choice Model. 4 (3), 52–74. Jan 1.

Kessels, R., Jones, B., Goos, P., 2015. An improved two-stage variance balance approach for constructing partial profile designs for discrete choice experiments. Appl. Stoch Model Bus. Ind. 31 (5), 626–648. Sep.

Louviere, J.J., Islam, T., Wasi, N., Street, D., Burgess, L., 2008. Designing discrete choice experiments: do optimal designs come at a price? J. Consum. Res. 35 (2), 360–375. Aug 1.

Luyten, J., Tubeuf, S., Kessels, R., 2022. Rationing of a scarce life-saving resource: public preferences for prioritizing COVID-19 vaccination. Health Econ. 31 (2), 342–362. Feb.

Maddala, T., Phillips, K.A., Reed Johnson, F., 2003. An experiment on simplifying conjoint analysis designs for measuring preferences. Health Econ. 12 (12), 1035–1047. Dec.

Meyerhoff, J., Oehlmann, M., 2023. The performance of full versus partial profile choice set designs in environmental valuation. Ecol. Econ. 204, 107665. Feb 1.

Mulhern, B., Norman, R., Shah, K., Bansback, N., Longworth, L., Viney, R., 2018. How should discrete choice experiments with duration choice sets be presented for the valuation of health states? Med. Decis. Making 38 (3), 306–318. Apr.

Nicolet, A., Perraudin, C., Wagner, J., Gilles, I., Krucien, N., Peytremann-Bridevaux, I., Marti, J., 2022. Patient and public preferences for coordinated care in Switzerland: development of a discrete choice experiment. Patient Patient Cent. Outcomes Res. 15 (4), 485–496. Jul.

Norman, R., Viney, R., Aaronson, N.K., Brazier, J.E., Cella, D., Costa, D.S., Fayers, P.M., Kemmler, G., Peacock, S., Pickard, A.S., Rowen, D., 2016. Using a discrete choice experiment to value the QLU-C10D: feasibility and sensitivity to presentation format. Qual. Life Res. 25, 637–649. Mar.

Norman, R., Mercieca-Bebber, R., Rowen, D., Brazier, J.E., Cella, D., Pickard, A.S., Street, D.J., Viney, R., Revicki, D., King, M.T., 2019. European organisation for research and treatment of cancer (EORTC) quality of life Group and the MAUCa consortium. UK utility weights for the EORTC qlu-C10D. Health Econ. 28 (12), 1385–1401. Dec.

Pickard, A.S., Law, E.H., Jiang, R., Pullenayegum, E., Shaw, J.W., Xie, F., Oppe, M., Boye, K.S., Chapman, R.H., Gong, C.L., Balch, A., 2019. United States valuation of EQ-5D-5L health states using an international protocol. Value Health 22 (8), 931–941. Aug 1.

Rose, J.M., Bliemer, M.C., 2013. Sample size requirements for stated choice experiments. Transportation 40, 1021–1041. Sep.

Rowen, D., Powell, P., Mukuria, C., Carlton, J., Norman, R., Brazier, J., 2021. Deriving a preference-based measure for people with Duchenne muscular dystrophy from the DMD-QoL. Value Health 24 (10), 1499–1510. Oct 1.

Sándor, Z., Wedel, M., 2005. Heterogeneous conjoint choice designs. J. Market. Res. 42 (2), 210–218. May.

Scarpa, R., Gilbride, T.J., Campbell, D., Hensher, D.A., 2009. Modelling attribute non-attendance in choice experiments for rural landscape valuation. Eur. Rev. Agric. Econ. 36 (2), 151–174. Jun 1.

Scherpenzeel, A., 2011. Data collection in a probability-based internet panel: how the LISS panel was built and how it can be used. Bulletin of Sociological Methodology/Bulletin de Méthodologie Sociologique 109 (1), 56–61. Jan.

Veldwijk, J., Marceta, S.M., Swait, J.D., Lipman, S.A., de Bekker-Grob, E.W., 2023. Taking the shortcut: simplifying heuristics in discrete choice experiments. Patient Patient Cent. Outcomes Res. 2, 1–5. May.

Welie, A.G., Gebretekle, G.B., Stolk, E., Mukuria, C., Krahn, M.D., Enquoselassie, F., Fenta, T.G., 2020. Valuing health state: an EQ-5D-5L value set for Ethiopians. Value in health regional issues 22, 7–14. Sep 1.